

AI Customer Support Chatbot – Project Requirements

ABC Technologies wants to develop an AI-powered customer support chatbot to automate customer interactions on their website and mobile application. The chatbot should be capable of understanding natural language queries, classifying customer requests, and providing accurate responses using backend APIs and knowledge base integration.

Project Objectives

- 1 Automate customer support responses using AI.
- 2 Reduce response time for customer queries.
- 3 Integrate chatbot with company backend systems.
- 4 Provide analytics dashboard for monitoring chatbot performance.

Key Features

- 1 Natural Language Understanding (NLU)
- 2 Customer query classification
- 3 Integration with backend APIs
- 4 Web-based dashboard for monitoring conversations
- 5 Training capability using historical support data

Suggested Technologies

- 1 Python for backend development
- 2 FastAPI for API services
- 3 PyTorch / Transformers for NLP model development
- 4 MongoDB for data storage
- 5 Docker for deployment

Project Constraints

- 1 The project should be completed within 45 days.
- 2 System must support at least 500 concurrent users.
- 3 Security and privacy of customer data must be maintained.